Titan Chronicles: Game Design & Architecture Blueprint

Working title. Assumes Attack on Titan as the base anime, built on a phone with GitHub Pages. This is the master plan every build session should follow.

1. Vision and Pillars

One-line pitch: Be born into the world of the walls, grow up through the major events of the story, and live them as a soldier: fast 3D maneuver combat, giant Titans, factions, and a world you can walk, ride and fly through.

Five pillars (every feature must serve at least one):

1. **Traversal is the fun.** ODM gear movement must feel fluid, fast and skillful. If this is not fun in a bare test arena, nothing else matters.
2. **Scale and dread.** Titans are huge, slow to read, deadly up close. The player feels small.
3. **You grow up.** Age, training, relationships and choices change who you are and which scenes you get.
4. **Canon moments as set pieces.** The big events happen around you, with your choices changing your role in them.
5. **Anime look.** Cel-shaded, bold outlines, dramatic camera, strong lighting and music.

2. Reality Check: What "Anime Quality" Means Here

Studio-animation quality (hand-drawn frames, film VFX) is not achievable solo in a mobile browser. What is achievable is an anime-style 3D look that reads as the anime:

- Cel shading (2 to 3 light bands), colored outlines, painted textures with little detail.
- Strong skyboxes, volumetric-looking fog, god rays, film-style color grading.
- Dynamic cinematic camera moments (slow-mo on kills, camera shake, speed lines, impact frames).
- Strong audio and music doing a lot of the emotional work.

Anime quality comes from art direction and presentation, not polygon count. If you already have realistic models and places, we can restyle them with the cel shader and outline pass so they match the anime look.

Engine decision

Option	Pros	Cons	Verdict
Three.js (3D, WebGL2)	ODM swinging needs 3D, runs on GitHub Pages, Claude writes it well	Mobile performance needs care	Chosen
Babylon.js	Built-in tools, physics	Heavier, less familiar	Backup
2D canvas / Phaser	Light, easy	Cannot capture ODM or Titan scale	Menus and minigames only
Unity / Unreal	Best quality	Cannot be built or run in a phone + GitHub flow	Rejected

Legal note

This is a fan project. Keep it non-commercial, use original or properly licensed assets only (no ripped models, music or screenshots from the anime), use an original player character, and use a working title that is not the anime's name.

3. Player Character and Life Structure

The player is an original character (custom name, appearance, starting home) placed beside the canon cast. This avoids replacing canon characters and gives freedom in choices.

Life stages

Stage	Age (approx.)	Location	What you do
1. Childhood	8 to 10	Shiganshina District	Walk-around life, friends, chores, town exploration, small quests, the day the wall breaks
2. Refugee years	10 to 12	Wall Rose interior, farmland	Survival, labor, choices about joining the military
3. Training Corps	12 to 15	Training grounds	Skill training, ODM tests, relationships, rankings, graduation
4. Garrison / Military Police / Survey Corps	15+	Trost, Capital, Wall Rose	Chosen faction determines missions and story access
5. War years	15 to 19	Across the walls	Expeditions, battles, uprisings, the return to Shiganshina
6. Outside world	late	Paradis, Marley	Final arcs

Stages 1 to 3 are the vertical slice (the first fully playable part). Everything after is chapters added over time.

Branching

At each stage the player picks a path that affects which missions, NPC bonds and combat roles are available. Canon events always happen, but your role in them varies (front line, support, medic, scout, command squad).

4. Core Gameplay Systems

4.1 ODM gear (the heart of the game)

- Two grapple hooks (left and right), each fired at a surface or Titan, aimed by camera crosshair with a soft auto-target assist for touch.
- Rope physics: pendulum-style swing using a simple spring/rope constraint, with reel-in to build speed and release to launch.
- Gas system: boosting, reeling and spinning consume gas. Gas refills at supply points (resource tension).
- Blades: blades dull and break with use, swapped from a limited carry count.
- Momentum model: speed is stored in the swing. Good players chain swings and keep speed, bad players run out of gas and fall.
- Camera: third-person chase camera that stretches FOV with speed.

Build this first in an empty test arena with towers and fake Titans. Tune it until it feels good before building anything else.

4.2 Combat

- Titan kill: hit the nape with a fast slashing pass. Hit angle and speed determine a clean kill versus a glancing hit. Limbs can be cut to cripple.
- Titan behaviors: each type uses a state machine (idle, wander, notice, chase, grab, attack, stagger, die).
- Human combat (later): blades vs. blades and thunder spears, mainly for Military Police and uprising scenes.
- Camera finishing moments: slow-mo plus camera swoop on a nape kill.
- Fatigue and fear: stress, low morale and injury affect aim and control, so panic becomes a gameplay factor.

4.3 Transformation (Shifter powers)

Shifter forms are a story-gated mechanic, not available from the start:

- Trigger by self-injury, with a cooldown and steam cost.
- Giant gameplay: slower, tankier, heavy attacks, area damage, a hardening ability, limited time and stamina.
- Each Nine Titan power is its own moveset. Start with one prototype Titan, add the others across chapters.

4.4 Progression

- Stats: Speed, Control, Strength, Endurance, Accuracy, Nerve, Tactics, Leadership.
- Skill trees: Mobility (swing efficiency), Blades (damage and durability), Support (medic/supply), Command (squad orders).
- Training grounds mini-games: balance training, ODM obstacle course, hand-to-hand, horse riding, strategy exams, all improving stats.
- Rank: cadet ranking affects branch choice at graduation (top ten may join the Military Police).

4.5 Relationships and morale

- Bond meters with key NPCs (friends, rivals, mentors).
- Dialogue choices alter bond and unlock scenes.
- Loss: characters can die. Canon deaths are fixed, but non-canon NPCs can die through player failure.

4.6 World systems

- Day/night cycle, weather, and time passing during peace phases.
- Horse riding on open terrain.
- Supply and gas economy between missions.
- Expedition formation (Survey Corps): long-range scouting formation with the player in a specific position, signal flares, and flanks collapsing when Titans arrive.
- Quest log, map, journal and relationship screen.

5. Story and Chapter Plan

Each chapter is a self-contained build target.

Chapter	Content	Core mechanic emphasis
0. Test Arena	Sandbox for ODM and Titan prototype	ODM feel
1. Shiganshina	Childhood walk-around, friends, town, the breach	Exploration, narrative, escape sequence
2. Refugee Years	Survival, labor, enlistment decision	Choice, tone
3. Training Corps	Training grounds, exams, graduation	Training mini-games, ODM tutorial
4. Trost	Fight to defend the district, supply HQ	First real combat chapter
5. Faction Path	Garrison / Police / Survey Corps missions	Faction branching
6. First Expedition	Forest chase, formation, losses	Large-scale combat, horses
7. Stohess / Capital	Urban chases, conspiracies	Urban ODM, stealth
8. Uprising and Return	Politics, battle for the walls	Squad commands
9. Outside World	Beyond the walls, Marley	New enemies and tech
10. Finale	Endgame set pieces	Giant combat

Scene types

- Interactive set pieces (full gameplay with objectives).
- Cinematic scenes (camera-directed with a few choices).
- Walk-and-talk (exploration and dialogue).
- Downtime (training, bonding, resting).

6. World and Maps

Each map is a self-contained scene with its own loading and a size budget. We do not build one giant open world; we stream zones and use loading screens and cinematic transitions.

Map	Description	Key features	Priority
Shiganshina District	Home town, near Wall Maria	Houses, market, canal, gate, wall climb	High (vertical slice)
Wall Maria (outer)	Giant wall seen from inside and top	Wall-top walkway, gate towers	Medium
Refugee Camps / Wall Rose inner	Farmland and shelters	Farms, camps, roads	Medium
Training Grounds	Cadet base	Barracks, obstacle courses, stables, mess hall, ODM tower	High (vertical slice)
Trost District	Gate-side city	Dense rooftops, HQ, supply depot, gate	High
Karanes District	Training city and supply base	Streets, barracks	Medium
Forest of Giant Trees	Huge trees as ODM anchors	Dense canopy, Titan chase zone	High
Utgard / Ragako Village	Ruined village	Interior spaces, set pieces	Low to medium
Underground City	Dark rough district	Tight tunnels, dim lighting	Medium
Stohess District	Urban, police presence	Rooftops, alleys, plaza	Medium
Mitras (Capital)	Royal capital	Grand buildings, cathedral	Medium
Wall Sina interior	Noble farmland	Estates, roads	Low
Shiganshina (ruins)	Destroyed Shiganshina	Damage states, ruins	Medium
Paradis Island overview	Island map	Travel hub	Low
Marley / Liberio	Outer world	Different architecture and tech	Later
Fort Slava / Odiha / Coastal	Late chapters	Battle zones	Later

Map build rules

- Each map ships as a modular kit (building pieces, walls, props) assembled in a level file (JSON), not a single giant mesh.
- Collision uses simplified proxy shapes.
- Roofs, walls and trees must be grappleable and tagged as anchor surfaces.
- Aim for 5 to 15 minutes of gameplay per zone, loading screens in between.

7. Factions and Characters

Factions

Faction	Role	Gameplay
Garrison Regiment	Wall defense	Defensive missions, stationed at gates
Military Police Brigade	City and king's guard	Urban missions, politics
Survey Corps	Exploration	Expeditions, high risk
Training Corps (cadets)	Student phase	Training, ranks
Royal Government	Political backdrop	Story context
Marley (late chapters)	Outside military	New enemies
Eldian resistance / Yeagerists (late)	Political forces	Uprising chapters

NPC tiers

- Tier 1 (canon leads, about 12): fully modeled, full expression set, custom animations.
- Tier 2 (supporting canon, about 25): unique model, reduced animations.
- Tier 3 (named side characters, about 40): kit-bashed from shared bodies with unique heads and uniforms.
- Tier 4 (crowd): 6 to 8 shared base bodies with color variants for civilians, cadets and soldiers.

8. Titans and Enemies

Pure Titans

Type	Size	Behavior
Small (3 to 5 m)	Low	Fast, erratic, lunge attacks
Medium (7 to 10 m)	Medium	Standard grab and bite
Large (13 to 15 m)	High	Slow, heavy hits, big grabs
Abnormal	Variable	Unpredictable, fast, dodges
Crawler	Variable	Low and fast, hard to hit nape
Sleeper	Variable	Dormant until approached

All Pure Titans share one rig with scaled variants and several unique head/body meshes. Behavior comes from the AI, not from unique animation sets.

Special enemies (Nine Titans and bosses)

Female, Armored, Colossal, Beast, Jaw, Cart, War Hammer, Attack, Founding. Each is a boss with a unique model, moveset, arena and cinematic intro. Add them one at a time after core gameplay is solid.

Human enemies

Marley soldiers, Military Police in uprising scenes, cultists, bandits. Simple human AI, with ranged and melee options.

9. Full Asset List

9.1 3D models

- Characters: 4 to 8 base body meshes, player customization (heads, hair, outfits, scarves), 12 Tier 1 characters, 25 Tier 2, 40 Tier 3 variants.
- Titans: 1 pure Titan rig and 8 to 12 head/body variants, 1 prototype shifter model first, then 8 more bosses.
- Gear: ODM gear (two versions), blades, thunder spears, gas tanks, cloaks, uniforms for each faction, horses (2 to 3 variants), carts, cannons.

- Environment kits: walls (stone, brick, wood), roofs, windows, doors, street props, market stalls, farms, tents, barracks furniture, crates, barrels, trees (8 species), rocks, grass, forest clutter, wall segments, gates.
- Landmarks: Wall Maria gate, Wall Rose gate, Trost HQ, training grounds tower, capital cathedral, Reiss chapel, Utgard castle.
- Destruction pieces: broken buildings, rubble, debris, wall chunks.

9.2 Textures and materials

- Painted, low-detail textures (anime style) at 512 to 1024 px, atlased.
- One shared cel-shader with outline pass.
- Material set: stone, wood, cloth, leather, metal, skin, grass, dirt, water, blood (stylized), steam.

9.3 Animation

- Player: idle, walk, run, jump, fall, land, crouch, climb, swim, interact, talk, sit, hurt, death, plus the entire ODM set (fire hook, reel, swing, boost, spin attack, blade strike, wall cling, landing roll), horse riding.
- Humans: shared set with variations (talk gestures, work, sit, cheer, grief, salute).
- Titans: idle, walk, run, crawl, notice, grab, bite, swat, stagger, fall, death steam, plus boss movesets.
- Cinematic: custom-posed scenes for key story moments.

9.4 VFX

- Hook rope lines, gas boost trails, steam, blood splash (stylized), impact frames, speed lines, dust and debris, fire, explosions, lightning for transformations, smoke, weather (rain, fog, snow), god rays.

9.5 UI and UX

- Touch controls: left virtual stick, right look area, buttons for hooks, boost, attack, jump, interact; optional gyro aim.
- HUD: gas, blades, health, stress, objective, minimap, compass.
- Menus: main, pause, map, journal, relationships, skills, inventory, settings (graphics quality), save/load.
- Dialogue box with portraits, choice UI, subtitle styles, chapter title cards.

9.6 Audio

- Music: original orchestral-style tracks per mood (calm, tension, battle, sadness, triumph), 12 to 20 tracks eventually.
- SFX: ODM sounds (hook fire, reel, gas hiss), blade hits, Titan footsteps and roars, ambient town and forest, UI sounds.
- Voice: optional later. Start with text and short vocal reactions.

9.7 Cinematics

- In-engine camera cutscenes (no pre-rendered video), letterbox bars, slow-mo, dialogue overlays.

9.8 Where assets come from (phone-friendly sourcing)

Source	Use	License notes
Models and places you already have	Characters, environments	Confirm you own or may use them
Kenney, Quaternius, Poly Pizza (CC0)	Base props, nature, buildings	Free, safe
Mixamo	Humanoid animations	Free with Adobe account, check terms

Source	Use	License notes
AI 3D generators (e.g. Meshy, Tripo)	Custom characters, props	Check each tool's terms
AI image tools	Concept art, textures, portraits	Original styles only
Freesound / OpenGameArt / original composition	SFX and music	Check licenses per file
Procedural generation in code	Terrain, trees, clutter, simple buildings	No license issue

Keep a license log: one line per asset with its source and license.

10. Technical Architecture

10.1 Stack

- Rendering: Three.js (WebGL2), custom cel-shader with outline pass, post-processing (bloom, color grade, vignette).
- Physics: Rapier (WASM) for collisions and rigid bodies, a custom rope/swing solver for ODM.
- Architecture pattern: lightweight ECS (entity-component-system) plus a game state machine.
- Audio: Web Audio API with sound pools and a music manager.
- Saves: IndexedDB for large saves, localStorage for settings.
- Hosting: GitHub Pages, static files only.
- Build: Vite for bundling and compressing, if a build step is possible; otherwise plain ES modules.

10.2 Repo structure

```
/
  index.html
  /src
    main.js                 (boot, loop)
    /core                   (engine, ecs, input, time, events)
    /render                 (renderer, cel-shader, post, camera)
    /physics                (world, odm-solver, collisions)
    /gameplay
      /player               (controller, odm, stats)
      /titans               (ai, types, bosses)
      /npc                  (dialogue, schedules, bonds)
      /combat
      /systems              (gas, blades, morale, time-of-day)
    /world                  (map loader, zones, streaming)
    /story                  (chapters, events, choices, flags)
    /ui                     (hud, menus, dialogue, touch-controls)
    /audio
    /save
  /assets
    /models  /textures  /anim  /audio  /ui
  /data                     (JSON: maps, npcs, quests, dialogue)
  /docs                     (this blueprint, change log)
```

10.3 Data-driven design

All maps, NPCs, quests, dialogue and Titan stats live in JSON files, not hard-coded. This lets us add content by editing data files, which is ideal for phone-based development.

10.4 Mobile performance budgets

Item	Target
Frame rate	30 fps minimum, 60 fps on high-end phones
Triangles on screen	150k to 300k
Draw calls	under 150 (use instancing and atlases)
Textures	512 to 1024 px, compressed where possible

Item	Target
Total download per chapter	under 60 MB
Titans on screen	3 to 6 active, rest are simple LODs
Shadows	one directional cascade, baked where possible

Quality settings: Low, Medium, High, with auto-detect.

10.5 Streaming and loading

- Zone-based loading with a loading screen or cinematic cover.
- glTF/GLB models, Draco or Meshopt compression, lazy-loaded.
- Preload next zone assets during calm moments.

10.6 GitHub Pages limits to respect

- Repository should stay well under 1 GB and individual files under 100 MB.
- Keep audio compressed (OGG/MP3 at modest bitrate).
- If assets grow too large, host heavy files on a separate static host and fetch them.

10.7 Phone-based workflow

1. Claude writes complete files or small patches per module.
2. You commit through the GitHub mobile app or web editor.
3. Test on the live GitHub Pages URL on your phone.
4. Report bugs with the console error text or a short description.

Rules: keep each file focused and small, one feature per session, and always keep the game runnable on the main branch.

11. Roadmap and Milestones

Milestone	Deliverable	Done when
M0: Test Arena	Cel-shaded scene, touch controls, ODM swing prototype, one dummy Titan	Swinging feels good on your phone
M1: Combat Prototype	Nape-hit kills, gas and blades, 3 Titan types, hit feedback	A fun 5-minute Titan fight
M2: Shiganshina Slice	Walkable town, 5 NPCs with dialogue, quest log, the breach and escape	A playable childhood chapter
M3: Training Corps	Training grounds, 4 mini-games, stats, ranking, graduation	Full cadet loop with saves
M4: Trost	First big combat chapter, squad AI	Playable battle with a story arc
M5: Faction Paths and Expedition	Faction choice, forest chase, formation	30 to 45 min of content
M6+: Later chapters	Stohess, uprising, outside world, bosses	Added chapter by chapter

Polish passes (lighting, color grading, audio, camera) run between milestones.

12. Risks and Mitigations

Risk	Mitigation
ODM feels bad	Prototype and tune first (M0), do not build content until fixed
Mobile performance	Strict budgets, quality settings, test on phone every build

Risk	Mitigation
Scope too big	Build chapter by chapter, vertical slice first
Asset quality inconsistent	One art bible, one shader, one palette
Asset licensing problems	Use only owned, CC0 or permitted assets, and keep a license log
Phone workflow limits	Small modules, data-driven content, simple repo structure
Copyright takedown	Non-commercial, original designs, no rips, no official names in the title

13. Decisions Needed

1. Confirm Attack on Titan as the base anime.
2. Confirm 3D (Three.js), the recommended choice for ODM and scale.
3. Player gender and look: fixed or customizable?
4. Tone: faithful dark and tragic, or slightly softer?
5. Time budget: how many hours per week can you build?
6. Do you have access to a PC or laptop at all? (affects asset pipeline options)
7. Which models and places do you already have, and what file formats are they in (GLB, FBX, OBJ)?

Once confirmed, the first build is M0: the Test Arena with working ODM swinging.